Team Control Number

15926

Problem Chosen

A

2024
HiMCM
Summary Sheet

Every four years, each Olympic Games sees the introduction, removal, or reintroduction of certain sports, disciplines or events (SDEs). Our team aims to determine whether a SDE will be included in the Olympics and developed a model scoring each SDE, while the score shows the potential of SDEs being selected.

In Task 1, we conducted factor analysis based on the six IOC criteria and identified 21 factors, along with their units and characteristics.

In Task 2, we used the 21 factors to construct the evaluation model. Given our data has relatively high dimensions while no assumptions on it, we adopted **Technique for Order Preference by Similarity to Ideal Solution (TOPSIS)** as our basic model, which aligns best with our data characteristics. To optimize model performance, we incorporated weight calculations into traditional TOPSIS method. First, we used the **entropy weight method (EWM)** to calculate the weights of each factor, and thereby obtain a score from TOPSIS integrating the weight. Next, we repeated the process using the **Analytical Hierarchy Process (AHP) method** to derive a second score. Finally, a weighted average of the two scores produces the final score, 0.6 for the first and 0.4 for the second. After the model is constructed, we confirmed its feasibility by inputting simulated data into it. Also, **we set a threshold of inclusion as 0.45,** based on the distribution of the simulation results

For Task 3, we selected three recently added SDEs (Sport Climbing, Skateboarding, and Surfing) and three long-standing SDEs (Swimming, Artistic Gymnastics, and Fencing) in the recent four Olympic Games as test subjects. After collecting data and inputting it into our model, we found new SDEs all scored above the given threshold 0.45, all **scoring high on Popularity and Accessibility principle**, especially Skateboarding; while the old SDEs were also steady above the threshold with a gradual increasing trend, **scoring similarly well on all 6 principles**. **These results align with real-world trends, demonstrating the generality of our model.**

In Task 4, we evaluated SDEs for future Olympics. Field Handball, Teqball, Rugby Union, and League of Legends were for 2032. By inputting estimated data into the model, we concluded that scores for Field Handball, Teqball, and League of Legends all indicated potential inclusion in 2032. The ranking was Teqball > League of Legends > Field Handball, with **Teqball scoring significantly high on Relevance and Innovation**. Next, we identified Baseball and Squash as potential candidates for the 2036 Olympics, performing similar process to compute their scores. **The model yielded scores above the threshold for both, suggesting they both have the potential of being selected.**

In Task 5, we conducted sensitivity analyses on both single and multiple factors. First, we performed single-variable analysis on the 21 factors and plotted the results, concluding that ***Continental Spread***, ***Injury Probability***, **and** ***F/M Athlete Ratio*** **were the most sensitive factors**. Then, we **examined intersection effects** between variables, selected two pairs of factors for multifactor analysis and plotted their results. Next, we conducted analysis for each of the six principles, **finding that** ***Inclusivity*** **was the most sensitive among them.** Lastly, since we used a weighted average method to calculate scores, we also performed a sensitivity analysis on the two weights used in the model.

Finally, we wrote a **letter to the IOC** presenting our model and findings.

Keywords: TOPSIS method, Entropy Weight method, Analytical Hierarchy Process method, Weighted Average method, Threshold of Inclusion, Single and Multiple Factor Sensitivity Analysis

Contents

1.Introduction 2
1.1. Background 2
1.2. Problem Restatement 2
2. Our Work 2
2.1. Assumptions and Justification 2
2.2. Key Factors Influencing the IOC Criteria (Task1) 3
2.2.1 Popularity and Accessibility 4
2.2.2 Gender Equity 4
2.2.3 Sustainability 4
2.2.4. Inclusivity 5
2.2.5. Relevance and Innovation 5
2.2.6. Safety and Fair Play 6
2.3. Evaluation Model for SDEs (Task2) 7
2.3.1. General Structure 7
2.3.2. Model part 1: TOPSIS 8
2.3.3. Model part 2: Entropy Weight Method 10
2.3.4. Model part 3: Analytic Hierarchy Process 11
2.3.5. Final Model 13
2.3.6. Simulation Test 14
2.4. Model Testing and Validation (Task 3) 14
2.4.1. New SDEs 14
2.4.2. Long-existing SDEs 15
2.4.3 Model Validation 16
2.5. Predicting Potential SDEs for the 2032 Olympics (Task 4) 16
2.5.1. Prediction for 2032 Olympics 16
2.5.2. Prediction for 2036 Olympics 17
2.6. Sensitivity Analysis (Task 5) 18
2.6.1. Single-factor Sensitivity Analysis 18
2.6.2. Multi-factor Sensitivity Analysis 19
2.6.3. Principle Sensitivity Analysis 20
2.6.4. Method-Weight Sensitivity Analysis 21
3. Strengths and Weaknesses 21
3.1. Strengths 21
3.2. Weaknesses 22
4. Letter to the IOC 23
5. References List 24
6. Appendix 24

1.Introduction

1.1. Background

The Olympic Games have long served as the world's most powerful symbol of athletic values and diversity. But as the dynamics of global sports and audience interest is constantly changing, the International Olympic Committee (IOC) had been continuously revising its evaluation process used to decide suitable sports, disciplines and events (SDEs), so that tournaments can match with the role. In the 2024 Paris Games, breaking made its debut, but the LA28 Games Plan also showed it would not be included in the upcoming Olympics. This obviously reflects how evaluation criteria changes rapidly, thus a sensible evaluation method that is able to accommodate the changes is crucial to a successful Olympics.

1.2. Problem Restatement

a) Identify features of SDEs, along with the six IOC criteria, **give choice reasons and include characteristics and units.**

b) Develop a model to **evaluate** which of the various SDEs align best with the current Olympic criteria.

c) Test model on 3 recently appeared SDEs and 3 long-existed SDEs. Then **compare the model's outcome with the actual result** and evaluate if the model aligns with it.

d) Predict 3 suitable SDEs that can be added or reintroduced to the 2032 Brisbane Olympics and **rate them first, second and third**. Then consider further beyond 2032.

e) Perform sensitivity analysis on the model and **identify which factors are most important to the decisions**, then evaluate whether the significance of these factors are advantageous or disadvantageous of the model.

f) Draft a **letter** addressed to the IOC reporting our conclusions.

2. Our Work

2.1. Assumptions and Justification

Assumption1: We only distinguish sex as male or female, based on physiological characteristics.

Justification： In all previous Olympic Games, athletes have been classified into the two categories based on physiological sex.

Assumption2: The SDEs are independent of each other and that their evaluations are mutually independent.

Justification： To simplify the model, we usually assume that the evaluations of different projects are mutually independent, thereby reducing the complexity within the model.

Assumption3: The popularity of the SDEs is closely related to the engagement of a global audience.

Justification： This assumption allows us to use existing audience data (such as television ratings) to quantify the popularity of SDE and to use this data to predict the project's long-term impact.

Assumption4: Assuming that time is not taken into account, each year represents an independent selection of sports events.

Justification： Each year, the inclusion of a particular sport in the Olympics will be considered independently.

2.2. Key Factors Influencing the IOC Criteria (Task1)

The factors that determine whether a sport can be featured as an Olympic event are numerous and often vary greatly, including aspects such as economic costs and social impact. Therefore, to ensure that we can comprehensively consider all factors that can determine whether any sport can be included in the Olympics, we consulted the official evaluation criteria set by the IOC. And based on the background provided in the topic, we ensured compliance with the six major criteria outlined in the topic. Furthermore, under the 6 criteria, we list a total of 21 more detailed factors as the variables for our model.

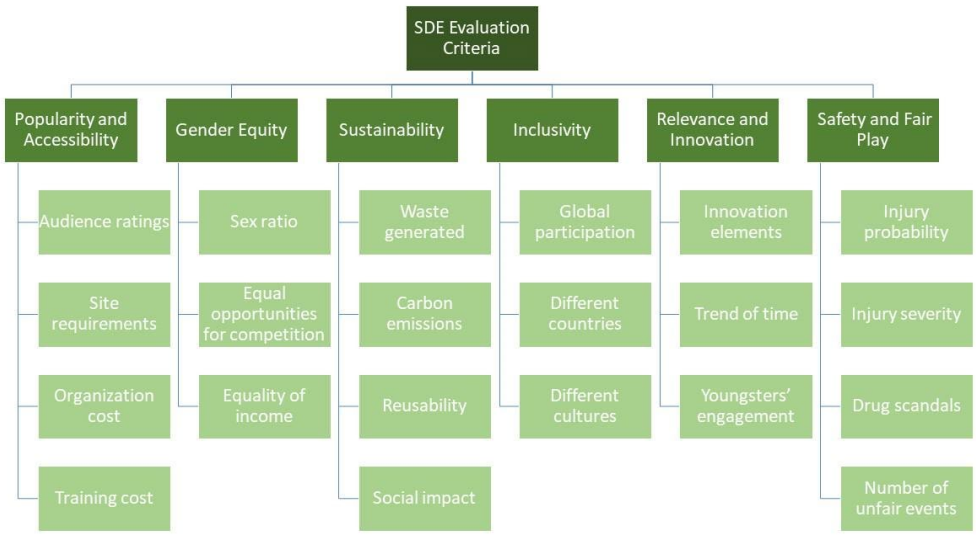


Figure1: Factors for IOC Criteria

2.2.1 Popularity and Accessibility

a) P: The average viewership rating of the sporting event per unit of time.

A quantitative, deterministic variable, unit in millions of viewers. Since the Olympics represent humanity, the number of people interested in a sport indicates its representativeness.

b) C1: The venue requirements for the sport.

A qualitative，probabilistic variable，no specific unit. Higher score indicates higher accessibility. SDEs with high venue requirements naturally forbids many participants from poorer regions.

c) C2: The cost required to host the sporting event.

A quantitative, deterministic variable, measured in million dollars. If the cost is excessively high, the participation of under-developed countries will be restricted.

d) C3: Entry requirements

A quantitative, deterministic variable measured in dollars/year. If a particular sport imposes high requirements regarding athletes' skills, funding, and facilities, it likely hinders participation, especially for athletes from poorer or developing regions.

2.2.2 Gender Equity

a)G_1: F/M Athlete Ratio

A quantitative, deterministic variable without units. One of the Olympic values is equality, and the closer the ratio of female to male athletes approaches is to 1, the more equal the gender ratio.

b)G_2: F/M Opportunity Ratio

A quantitative, deterministic variable without units. It examines whether equal competition opportunities are provided for both male and female athletes in the event.

c)G_3：F/M Income Ratio

A quantitative, deterministic variable without units. The unbalanced income of male and female athletes is a pressing issue that needs to be addressed to achieve equity.

2.2.3 Sustainability

a)W_1：The waste generated during the production process of the facilities and venues required for the SDEs.

A quantitative, deterministic variable measured in tons. Since most SDEs require special venues, there will be waste during industrial production which causes environmental impact of the SDE.

b)W_2：The carbon emissions resulting from hosting the events.

A quantitative, deterministic variable, measured in tons of carbon dioxide. Carbon dioxide is emitted during the production of the facilities, transportation to deliver competition equipment, and maintenance, which leads to significant carbon emissions while carbon neutrality is an important goal in today's world.

c)R: The degree of reusability of the facilities and venues prepared for the project.

A qualitative, probabilistic variable, no units. Olympics usually need to temporarily construct many venues, but if the degree of reusability is high, there will be less waste of resources.

d)SI: Social Impact.

A qualitative, probabilistic variable, with no units. Olympics promote peace among people globally, therefore, it is essential to eliminate elements that may leave a negative impression regarding social peace.

2.2.4. Inclusivity

a)N_1: Whether the SDEs is likely to promote global participation.

A qualitative, deterministic variable, with no units. IOC clearly states that a sport must have at least 75 countries actively participating to qualify for inclusion.

b)N_2: The number of countries participating in the sport event.
A quantitative, deterministic variable, unit is count of countries. Building on the previous point, the more countries participating actively, the more diverse and inclusive the sport is considered to be.

c) N_3: The number of countries from different cultural backgrounds among the participating nations.
A quantitative, deterministic variable, unit is the count of continents. Building on the previous point, if the participating countries share similar cultures, even a large number of countries would not contribute significantly to diversity.

2.2.5. Relevance and Innovation

a)T_1: Whether the sport can incorporate new technologies.
A qualitative, probabilistic variable, with no unit. The Olympics should also showcase the progress of human technological development, we can analyze whether a given sport has innovativeness and demonstrates the latest technological achievements.

b)T_2: Whether the sport aligns with contemporary trends.
A qualitative, probabilistic variable, with no unit. We can analyze whether a SDE is related to recent trending topics and suggest if it reflects current trends and indicates its popularity among people.

c)YV: The proportion of young viewers among the total audience.
A quantitative, deterministic variable, with no unit. Since young viewers are generally

more open to new and innovative sports, a higher proportion of young viewers in the total audience indicates greater innovation and relevance of the sport.

2.2.6. Safety and Fair Play

a)I_1: The probability of athlete injuries.
A quantitative, deterministic variable, with no unit. As one of the most influential international competitions, the Olympics must strictly consider the safety of athletes. The probability of injury for each event reflects the safety of a SDE.

b)I_2: The severity of athlete injuries.
A qualitative, probabilistic variable, with no unit. By combining the analysis from the previous point with the average severity of past injury cases for each event, we can assess the safety of each sport more wholly.

c)D: The number of times athletes have been found using prohibited substances.
A quantitative, deterministic variable, measured in counts of people. To quantify the fairness of event regulations, we can deduct from the historical instances of such scandals for each sport to reveal the fairness of that sport.

d)F: Fairness level.
A qualitative, probabilistic variable. Based on the number of such news reports and the judgments regarding the fairness of the rules, this can become our final factor in determining the fairness of each sport.

Table 1: Factors for IOC Criteria

Principle	Symbol	Definition	Unit	Characteristics
Popularity and Accessibility	P	Average viewership of the sports	million people	Deterministic, Quantitative
	C_1	Whether the venue requirements for the sport are high	N/A	Probabilistic, Qualitative
	C_2	Cost of setting up the venue required for the sport	million dollar	Deterministic, Quantitative
	C_3	Training costs for athletes to reach competition level	dollar/year	Deterministic, Quantitative
Gender Equity	G_1	Ratio of female to male athletes	N/A	Deterministic, Quantitative
	G_2	Ratio of female to male competition opportunities	N/A	Deterministic, Quantitative
	G_3	Average income ratio of female to male athletes in events	N/A	Deterministic, Quantitative
Sustainability	W_1	Waste generated from setting up facilities and venues for the sport	tons	Deterministic, Quantitative

	W_2	Carbon dioxide emissions from hosting the event	tons	Deterministic, Quantitative
	R	Degree of reuse of the facilities required for the sport	N/A	Probabilistic, Qualitative
	SI	Whether the social impact is positive	N/A	Probabilistic, Qualitative
	N_1	Whether more than 75 countries participate in this sport	N/A	Deterministic, Qualitative
Inclusivity	N_2	Number of countries participating in this event	countries	Deterministic, Quantitative
	N_3	Number of different cultural regions represented by participating countries	continents	Deterministic, Quantitative
	T_1	Whether the sport can incorporate new technologies	N/A	Probabilistic, Qualitative
Relevance and Innovation	T_2	Alignment of the sport with current trends	N/A	Probabilistic, Qualitative
	YV	Proportion of young viewers among total viewers	N/A	Deterministic, Quantitative
	I_1	Probability of athletes getting injured while playing the sport	N/A	Deterministic, Quantitative
	I_2	Severity of injuries if athletes are injured	N/A	Probabilistic, Qualitative
Safety and Fair Play	D	Number of times athletes test positive for banned drugs	people	Deterministic, Quantitative
	F	Fairness of the scoring standards	N/A	Probabilistic, Qualitative

Overall, among the 21 factors, there is no constant and this attribute is not explicitly listed in the table. 7 factors that are probabilistic, they are scores obtained from subjective integration and judgment; thus, they are all qualitative. In practice, these scores with relatively low accuracy should be replaced by more authoritative expert assessments to ensure the model's accuracy. Among the remaining 14 deterministic factors, only one, Global Reach, is qualitative, while the others are directly derived from objective data. Together, these 21 factors represent what we believe are the most important indicators in determining whether a sport can be included in the Olympic Games. And for collecting data for the factors, references [2]-[10] are our main sources.

2.3. Evaluation Model for SDEs (Task2)

2.3.1. General Structure

In order to evaluate whether a SDE can be included in the Olympics, and considering that we have collected data on 21 factors, which is quite a lot, we conclude a multi-

dimensional evaluation problem. Therefore, we adopted the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) method as our basic model for the following reasons:

1. Multi-Dimensional Evaluation

TOPSIS considers multi-dimensional data by defining the positive ideal solution (the optimal scenario) and the negative ideal solution (the worst scenario), and evaluate SDEs based on its distance from these two ideal solutions.

2. High Interpretability

TOPSIS provides good interpretability, which is particularly necessary for the Olympics to everyone interested in the games why certain projects were included or excluded.

3. Low Data Requirements

TOPSIS does not require any assumptions or statistical analysis of the data, and only requires the weights of each attribute and the data of various factors.

The structure of the complete model is shown in the figure below:

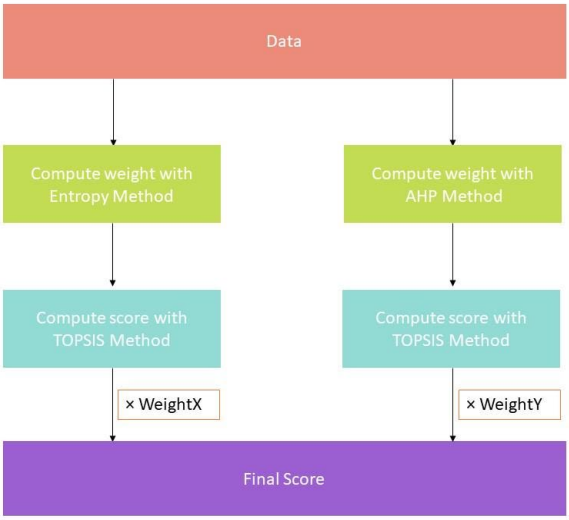


Figure 2：Model Structure

2.3.2. Model part 1: TOPSIS

Based on the reasons mentioned in Section 2.3.1, for task 2, we applied the TOPSIS method to calculate the specific scores of different sports.

First, we transformed the listed factors to ensure that all factors followed the trend of “the larger, the better.” For indicators that were originally "the smaller, the better", the following formula was used to convert them into maximizing indicators:

$$M = \frac{1}{x}$$

For indicators originally "the closer to a certain value, the better", the following formula was used to convert them into maximizing indicators, where x_{best} represents the desired value.

$$M = max\{|x_i - x_{\text{best}}|\}, \bar{x}_i = 1 - \frac{|x_i - x_{\text{best}}|}{M}$$

After transformation, we performed standardization on the data. The purpose is to eliminate differences between features and the impact of different scales. Here, we constructed a normalized matrix containing n evaluation objects and 21 evaluation indicators (i.e., an n times m =21 matrix), incorporating the objects and indicators into the following matrix:

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1m} \\ x_{21} & x_{22} & \cdots & x_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nm} \end{bmatrix}$$

The standardized matrix is denoted as Z . Each element of Z is processed as follows:

$$z_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^{n} x_{ij}^2}}$$

The standardized matrix Z is then obtained:

$$Z = \begin{bmatrix} z_{11} & z_{12} & \cdots & z_{1m} \\ z_{21} & z_{22} & \cdots & z_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ z_{n1} & z_{n2} & \cdots & z_{nm} \end{bmatrix}$$

Next, the largest value from each column of matrix Z is selected and defined as Z^+ , as expressed by the following formula:

$$Z^+ = (max\{z_{11}, z_{21}, \cdots, z_{n1}\}, max\{z_{12}, z_{22}, \cdots, z_{n2}\}, \cdots, max\{z_{1m}, z_{2m}, \cdots, z_{nm}\})$$

Then, the smallest value from each column of matrix Z is selected and defined as Z^- , as expressed by the following formula:

$$Z^- = (min\{z_{11}, z_{21}, \cdots, z_{n1}\}, min\{z_{12}, z_{22}, \cdots, z_{n2}\}, \cdots, min\{z_{1m}, z_{2m}, \cdots, z_{nm}\})$$

After that, the distance D_i^+ between each evaluation indicator in a row and the corresponding maximum value is calculated using the following formula, and i (i = 1, 2, 3, ... , n) is defined as：

$$D_i^+ = \sqrt{\sum_{j=1}^{m} \left(Z_j^+ - z_{ij}\right)^2}$$

Next, the distance D_i^- between each evaluation indicator in a row and the corresponding minimum value is calculated using the following formula, and i (i = 1, 2, 3, ... , n) is defined as:

$$D_i^- = \sqrt{\sum_{j=1}^{m} \left(Z_j^- - z_{ij}\right)^2}$$

However, when considering the TOPSIS method, an issue occurred: we assumed that all factors contribute equally to the final evaluation, which is unrealistic. To further optimize the model and make it more aligned with reality, we revised the formulas for D_i^+ and D_i^- to the following, where ω_j represents the weight of factor j:

$$D_i^+ = \sqrt{\sum_{j=1}^{m} \omega_j \left(Z_j^+ - z_{ij}\right)^2}, D_i^- = \sqrt{\sum_{j=1}^{m} \omega_j \left(Z_j^- - z_{ij}\right)^2}$$

Next, we introduced other 2 methods to determine the weight of each factor.

2.3.3. Model part 2: Entropy Weight Method

This method calculates the information entropy of each indicator based on the degree of dispersion in the values of the indicators, and thereby determines the importance of each indicator (i.e., the weight). With greater degree of dispersion of an indicator, the larger the weight.

The specific calculation method is as follows:

The first step is to calculate the value proportion Pij of the j-th factor for the i-th SDE:

$$P_{ij} = \frac{x_{ij}}{\sum_{i=1}^{n} x_{ij}} (i = 1,2 \dots, n; j = 1,2 \dots, m)$$

Then, the value of k for the matrix is calculated based on n (the number of rows in the matrix):

$$k = \frac{1}{\ln(n)}$$

Based on this, the entropy value e_j of the j -th factor is further calculated:

$$e_j = -k * \sum_{i=1}^{n} P_{ij} * \ln\left(P_{ij}\right)$$

Where $k \geq 0$ and $e_j \geq 0$.

Next, the variation index d_j of the indicator is calculated:

$$d_j = 1 - e_j (j = 1,2 \dots, m)$$

Finally, the entropy weight w_j of the indicator is calculated as:

$$w_j = \frac{d_j}{\sum_{j=1}^{m} d_j}$$

By applying these steps to a matrix containing 21 indicators and n evaluation items, the weight of each indicator can be determined.

However, we found that relying solely on EWM has some limitations:

1. Some factors rely on subjective ratings of assigned scores. As a result, the degree of dispersion for these factors are all small, leading to small weights for them that does not fit reality.

2. Some variables, such as whether participating countries exceed 75, are prerequisites so all SDEs have the same value of them. And the entropy weights of these factors become 0. Although we introduced a small perturbation in computation to prevent their weights from being exactly 0, but these weights are still very small.

So we concluded that another different model is needed to calculate the weights. And AHP (Analytic Hierarchy Process), a method based on subjective ratings to calculate weights, the opposite of entropy weight method, was the solution we chose.

2.3.4. Model part 3: Analytic Hierarchy Process

The APH model was first developed by Thomas Saaty in the 1970s. It handles the hierarchical structure of variables and can calculate the weight of each variable. The hierarchy structure is as follows:

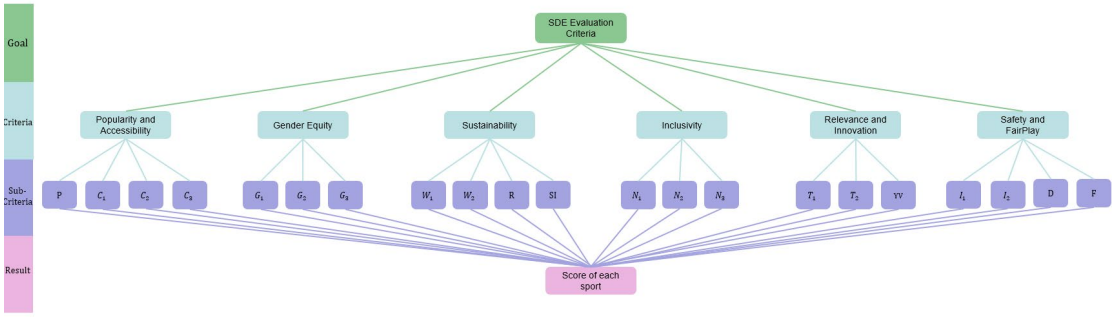


Figure 3: AHP Hierarchy Structure

Table 2：Scale Table

Scale	Definition
1	factor i is equally important as factor j
3	factor i is slightly more important than factor j
5	factor i is apparently more important than factor j
7	factor i is strongly more important than factor j
9	factor i is extremely more important than factor j
2,4,6,8	the intermediate values
reciprocal	importance scale between factor j and factor i

According to the 1-9 scale, a comparison matrix can be established to evaluate the importance of various variables. This matrix is represented as a square matrix, where each entry C_{ij} signifies the importance of variable C_j in relation to variable C_i. The values in the matrix are derived from a predetermined scale, as illustrated in the

table above. Through this structured approach, we can quantitatively assess and compare the significance of different variables in the context of our analysis.

$$C = \begin{bmatrix} C & C_1 & C_2 & C_3 & \dots & C_n \\ C_1 & C_{11} & C_{12} & C_{13} & \dots & C_{1n} \\ C_2 & C_{21} & C_{22} & C_{23} & \dots & C_{2n} \\ C_3 & C_{31} & C_{32} & C_{33} & \dots & C_{3n} \\ \dots & \dots & \dots & \dots & \dots & \dots \\ C_n & C_{n1} & C_{n2} & C_{n3} & \dots & C_{nn} \end{bmatrix}$$

To assess the importance of variables in the model, we primarily calculated the weights of the 6 principles, as shown below:

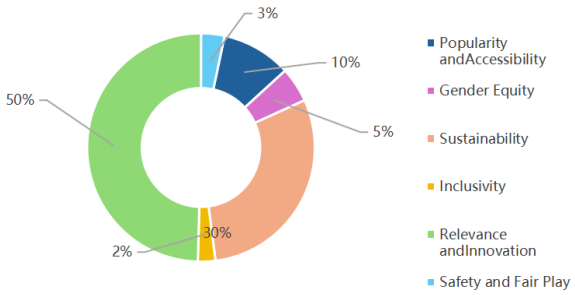


Figure 4: Principle Weight Pie Chart

We also calculated the weights for each factors. The following are the results summarized based on preferences for the Olympics.

Table 3: AHP Weights

Destination Layer	Principle layer	Weight-1	Sub-Criterion Layer	Weight-2	Final Weight
SDEs to include in the Olympics	Popularity and Accessibility	0.0991	P	0.4285	0.0425
			C_1	0.2571	0.0255
			C_2	0.1714	0.0170
			C_3	0.1428	0.0141
	Gender Equity	0.0495	G_1	0.1666	0.0082
			G_2	0.5000	0.0247
			G_3	0.3333	0.0165
	Sustainability	0.2975	W_1	0.2790	0.0830
			W_2	0.0930	0.0276
			R	0.0697	0.0207
			SI	0.5581	0.1660
	Inclusivity	0.0247	N_1	0.7741	0.0191
			N_2	0.1290	0.0031
			N_3	0.0967	0.0023
		0.4958	T_1	0.1250	0.0619

Relevance and Innovation		T_2	0.1250	0.1859
		YV	0.5000	0.2479
		I_1	0.1714	0.0056
Safety and Fair Play	0.0330	I_2	0.0857	0.0028
		D	0.5142	0.0170
		F	0.2285	0.0075

Note：weight-1 is the weight proportion of the criterion level, and weight-2 is the weight proportion of the objective level

In addition, to verify the consistency of our assumed inputs and thereby support the realism of our fictional characters, we examined the consistency of the n times n matrix, whose maximum eigenvalue λ max is the largest eigenvalue of the matrix. We can express this eigenvalue using the following formula

$$\mathrm{CI} = \frac{\lambda_{max} - n}{n - 1}$$

After calculating Cl, we further introduced RI and CR to determine if the level of inconsistency stays within the tolerance range of 0.1.

$$\mathrm{CR} = \frac{\mathrm{Cl}}{\mathrm{RI}} < 0.1$$

Table 4: Random Consistency Index

n	3	4	5	6	7	8	9
RI	0.58	0.90	1.12	1.24	1.32	1.41	1.41

$$CR = \frac{CI}{RI} < 0.1$$

After calculations, we confirmed that the CR is always below 0.1, which means our comparison matrix is realistic and consistent.

Finally, we can calculate the score of the i-th evaluation object (i = 1, 2, 3, ..., n) using the following formula

$$S_i = \frac{D_i^-}{D_i^+ + D_i^-}$$

And it can be concluded that $0 \leq S_i \leq 1$.

2.3.5. Final Model

As we used two methods to calculate the weights of each factor, we used weighted average method to integrate them. Let the score obtained using the weight W_{ai} (derived from the AHP method) be S_{ai} , and let the score obtained using the weight W_{ei} (derived from the entropy weight method) be S_{ei} . Finally, the weighted average of the two scores, S_{fi} , represents the final score (where the weight of S_{ai} is W_x and the weight of S_{ei} is W_y), as shown in the following formula:

$$S_{fi} = W_x \times S_{ai} + W_y \times S_{ei}$$

The calculation of the two weights is determined by the proportion of objective and subjective data in the total data.

2.3.6. Simulation Test

After constructing our model, we tested it using a set of simulated data containing 8 SDEs.

Although the theoretical score range of our model is between 0 and 1, we specifically set up 8 data sets with significant differences to test the distribution of scores when specific data is applied. In order to determine a score threshold for including an SDE. Additionally, we further optimized the data for the highest-scoring items under realistic conditions, ensuring that it could reflect the potential highest score achievable by the model.

The final results obtained are as follows:

Table 5: Simulation Scores

SDE A	0.439471	SDE C	0.364057	SDE E	0.136761	SDE G	0.499521
SDE B	0.782946	SDE D	0.411277	SDE F	0.428866	SDE H	0.331539

From the table, we find that the scores obtained from the model actually range between 0 and 0.8, with most scores clustering between 0.3 and 0.5. Therefore, we set the minimum score required for a project to pass the evaluation, as 0.45. This score is closer to the distribution of most of the data, making it a suitable threshold.

2.4. Model Testing and Validation (Task 3)

According to Task 3, next we should test the model constructed. In accordance with the requirements of the task, we selected the following six sports:

1.Sports Added or Removed in the Last Four Olympic Games:

Rock Climbing, Skateboarding, Surfing.

2. Sports that Have Continuously Appeared Since the 1988 Olympics or Earlier:

Swimming, Artistic Gymnastics, Fencing.

To ensure the generality of the model, during the testing process, for any selected sport, we will collect relevant data for all 21 factors separately and then plug the collected data into the model for calculation. Additionally, the SDEs chosen for testing are all sports with significant differences.

2.4.1. New SDEs

First, we tested the three recently added sports. We collected, transformed, and standardized the data based on the 21 factors. During the testing process, we input the collected data into the model for calculation and obtained the following scores:

Table 6: New SDEs Final Scores

SDE	TE_Score	TA_Score	Final_Score
Rock Climbing	0.5743	0.5071	0.5474
Skateboarding	0.5253	0.6292	0.5669
Surfing	0.5539	0.5164	0.5389

Note: TE_Score represents the score obtained from the TOPSIS model using weights calculated by the entropy weight method. TA_Score represents the score calculated using weights obtained from the AHP method, while Final_Score is the final score obtained by weighted averaging the two previous scores.

Their scores at the Principle level are as follows:

Table 7: New SDEs Principle Scores

Principle Score	**Popularity & Accessibility**	**Gender Equity**	**Sustainability**	**Inclusivity**	**Relevance and Innovation**	**Safety and Fair Play**
Rock Climbing	0.4471	0.4957	0.4273	0.1732	0.2916	0.5065
Skateboarding	0.6080	0.3439	0.3990	0.2508	0.4047	0.3549
Surfing	0.3897	0.4584	0.5757	0.2206	0.2750	0.4269

It can be seen that these projects not only exceeded the standard score line in the selection process of the recent Olympic Games, but are also relatively high. This aligns with the fact that they were selected as new events in the latest editions of the Olympics.

2.4.2. Long-existing SDEs

Secondly, we collected data on the long-existing SDEs in 2000, 2012, and 2024. Then calculated the comprehensive scores for them and created the following chart:

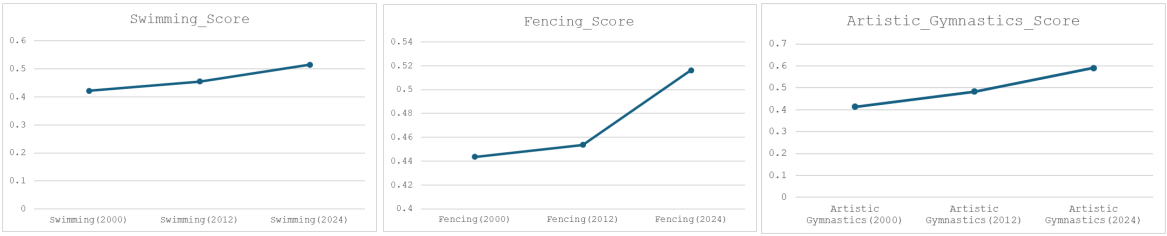


Figure 5: Old SDEs Final Scores over 2000-2024

The scores at their principle level and the final scores are as follows:

Table 8: Old SDEs Principle Scores

Principle Score	Popularity & Accessibility	Gender Equity	Sustainability	Inclusivity	Relevance and Innovation	Safety and Fair Play
Swimming	1.6488	1.9668	2.5368	2.0000	1.7800	3.0000
Artistic Gymnastics	2.1948	0.1610	2.6306	0.6436	2.8800	0.3912
Fencing	2.6569	2.4264	3.9996	1.8210	0.8000	3.7763

Table 9: Old SDEs Final Scores

Sport	TE_Score	TA_Score	Final_Score
Swimming	0.4956	0.4620	0.4821

Artistic Gymnastics	0.4202	0.5990	0.4917
Fencing	0.6467	0.5063	0.5906

If any sport has been included in multiple editions of the Olympics, it should have relatively stable and upward-trending scores. Based on the scores, we can conclude that their scores are lower than those of the top three newly added events, yet still above our selection criterion of 0.45. Furthermore, their scores have remained stable over several years, which supports their continued presence in the Olympics.

2.4.3 Model Validation

Based on the tests conducted on the two sets of sports, the magnitude and trend of the scores both aligns with the results presented in the given table. This demonstrates that the model is applicable to any type of sport, allowing for accurate assessments of whether a sport is included in the Olympics.

2.5. Predicting Potential SDEs for the 2032 Olympics (Task 4)

2.5.1. Prediction for 2032 Olympics

For task 4, we first analyzed which sports could be newly added or reintroduced for the 2032 Brisbane Olympics, selecting the following three sports for scoring purposes:

1.Field Handball - Outdoor handball debuted in 1936, while indoor handball has been in the Olympics since 1972. With growing interest in outdoor sports and lower infrastructure needs, especially post-pandemic, outdoor handball aligns with the 2032 Brisbane Olympics' focus and its popularity in Australia supports its future inclusion.

2.Teqball - Teqball, a fast-paced sport based on football and played on a table similar to table tennis, has never been part of the Olympics. Its popularity among Gen Z, coupled with low venue construction costs, makes it a strong contender for future Olympic inclusion.

3.Rugby Union - Rugby Union, previously in the Olympics before 1924, is a well-established sport with a strong fan base. With Rugby Sevens in the Games since 2016, it deserves a chance for innovation. Australia's rugby strength could attract Rugby World Cup fans, promoting global participation and innovation, making it a fitting Olympic sport.

4.League of Legends - League of Legends, a popular e-sports, has high viewership, especially among younger audiences. With low participation requirements and complex gameplay, it challenges players' skills and resilience. As a team sport, it embodies the Olympic spirit, making it a strong candidate for inclusion.

Therefore, we collected data on the 21 influencing factors related to these four sports and performed calculations using our model, resulting in the following scores for the three sports:

Table 10: 2032 SDEs Final Scores

Sport	TE_Score	TA_Score	Final_Score
Field Handball	0.5095	0.4698	0.4936

Teqball	0.6874	0.6217	0.6611
Rugby Union	0.3318	0.3211	0.3275
League of Legends	0.5190	0.6888	0.5869

And their scores on each principle level are as follows:

Table 11: 2032 SDEs Principle Scores

Principle Score	**Popularity & Accessibility**	**Gender Equity**	**Sustainability**	**Inclusivity**	**Relevance and Innovation**	**Safety and Fair Play**
Field Handball	0.3837	0.3194	0.7647	0.0461	0.0214	0.4040
Teqball	0.3573	0.2811	0.5356	0.1582	0.3215	0.6147
Rugby Union	0.1586	0.5470	0.4583	0.6416	0.0687	0.2212
League of Legends	0.3779	0.1911	0.5057	0.2939	0.5057	0.4718

After obtaining the scores, we ranked and analyzed the scores of the four sports, leading to the following conclusions:

Teqball is the most likely sport among the four to be included in the Brisbane Olympics, followed by League of Legends, and lastly Field Handball. Rugby Union scored below 0.45, the selection threshold, therefore unlikely to be included.

2.5.2. Prediction for 2036 Olympics

Then, we conducted an analysis of the sports that may be added to the 2036 Olympics, and we believe the following two sports have a high likelihood of being included:

Baseball: Baseball, known for its rich history and global popularity, is a highly competitive sport requiring skill and strategy. It appeals to younger audiences, with many schools offering baseball courses, and will return to the Olympics in 2028. These factors suggest that baseball is likely to be included in the 2036 Olympics.

Squash: Squash is a competitive sport with global participation, requiring speed, endurance, reflexes, and strategy. It promotes gender equality and is played indoors, unaffected by weather. These factors suggest squash is likely to be included in the 2036 Olympics.

Next, following the same procedure as before, we collected data on 21 influencing factors related to these two sports, transformation and standardized them, and finally incorporated them into our model for calculation, resulting in the following scores:

Table 12: 2036 SDEs Final Scores

Sport	TE_Score	TA_Score	Final_Score
Baseball	0.4370	0.4781	0.4535
Squash	0.6570	0.4657	0.5805

In terms of the Principle layer, their scores are as follows:

Table 13: 2036 SDEs Principle Scores

Principle Score	Popularity & Accessibility	Gender Equity	Sustainability	Inclusivity	Relevance and Innovation	Safety and Fair Play
Baseball	0.4693	0.1564	0.5738	0.5738	0.2816	0.1317
Squash	0.3207	0.3565	0.5284	0.5284	0.1184	0.4443

The final scores for both sports are above 0.45, which meets the selection criteria. However, there is a clear difference in the scores; it can be observed that squash has a greater likelihood of being included in the Olympics. On the other hand, baseball's score is only slightly above the qualifying line by just 0.003 points. Therefore, although it meets the qualification requirements, it is highly likely that due to the disparity in data, its score could fall below the qualifying line, resulting in its exclusion.

2.6. Sensitivity Analysis (Task 5)

2.6.1. Single-factor Sensitivity Analysis

After testing the model, we conducted a sensitivity analysis on the 21 factors within the model. We divided the analysis into multivariable and univariable forms. First, we analyzed each of the 21 factors mentioned above individually using controlled variable method.

However, during analysis, we found that because we took into account the real meaning of each variable when setting the single-variable ranges, the scales of the various variables are inconsistent. Therefore, it is impossible to directly judge their sensitivity based on the slopes of the graphs. Hence, their sensitivities cannot be directly compared from the graphs.

To address this issue, we subsequently standardized all the variables to ensure they are on the same scale and then re-plotted the graph, as shown below:

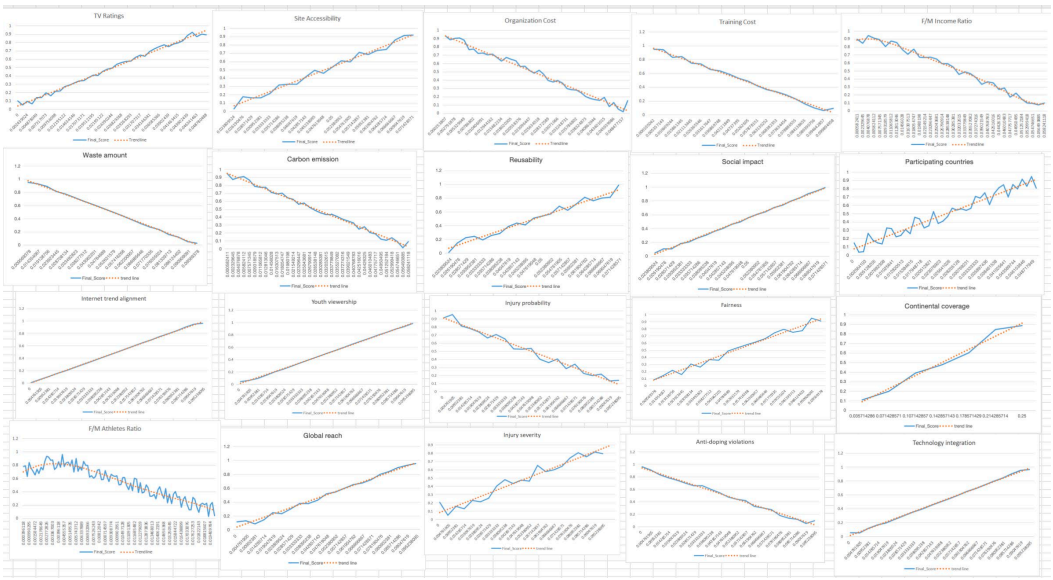


Figure 6: Single-Factor Sensitivity Analysis

From the graph (one graph is excluded to keep the shape and is attached in the appendix), it can be seen that our 21 factors can be divided into three categories: The larger the value, the higher the score (such as event viewership); The smaller the value, the higher the score (such as training costs); The closer the value is to a certain target, the higher the score (such as the male-to-female athlete ratio).

Among all the indicators, we can observe from the slope of the fitted line in the graph that the importance of the indicators is relatively similar. However, the few indicators that have a relatively larger impact on the final score are:

1.Continental Spread, which refers to the cultural diversity represented by the countries participating in the sport, is such that the larger the value, the higher the score.

2.Injury Probability, which refers to the likelihood of athletes participating in the sport getting injured, is such that the lower the value, the higher the score.

3.F/M Athlete Ratio, which refers to the ratio of female to male athletes participating in the sport, is such that the closer the value is to 1, the higher the score, while the impact on the final score decreases.

These three indicators align with the principles recently promoted by the IOC, namely global participation, safety, and gender equality. Sports that score high on these three indicators and are thus selected for the Olympics can naturally help promote these Olympic values.

2.6.2. Multi-factor Sensitivity Analysis

After conducting an independent sensitivity analysis on the 21 variables, we realized that there may be certain influences and relationships between the variables. Since the scores of some variables increase as their values increase, while others behave oppositely, the graphs of such variables are likely to have at least one intersection. This reflects the possibility of a joint effect, thus impacting the final score accordingly. By observing the graphs, we identified two pairs of indicators that may have such relationships: Continental Coverage and F/M Athlete Ratio; Fairness and Anti-doping violations. The final results are as follows:

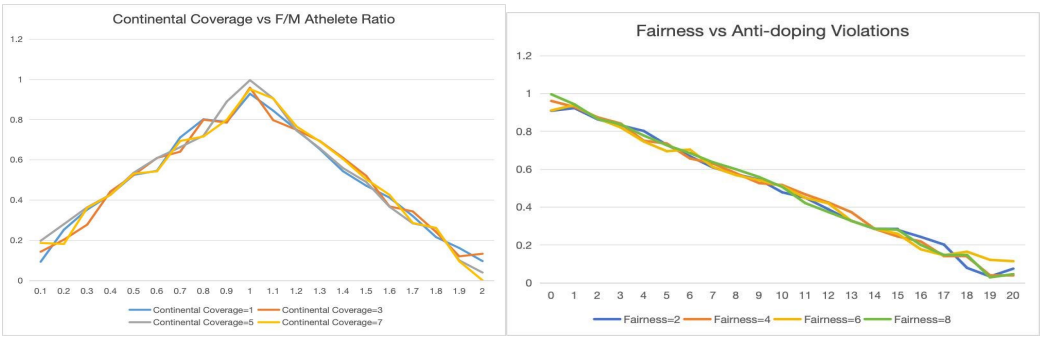


Figure 7: Multi-Factor Sensitivity Analysis

The graph on the left shows that when Continental Coverage is 5, the highest score is significantly higher than in the other groups, very close to 1. We believe this is because developed countries, which are more focused on and advocate for gender equality, tend to be concentrated in certain continents (such as Europe), while other continents (like Africa) are relatively underdeveloped and pay less attention to gender

equality in sports. As a result, a sport that includes all continents may actually have lower gender equality. The results of this sensitivity analysis demonstrate the model's ability to reflect reality. However, since it provides similar results across other data groups as well, this ability is clearly limited.

When analyzing Fairness and Anti-Doping Violations together, although the differences are not large, it can still be seen in the graph on the right that when Fairness is at its highest, 8, the overall score is higher than in the other data groups. We believe this is due to the direct link between the clarity and strictness of the competition format and the level of restriction on athletes' use of banned substances. Similarly, this also reflects the model's ability to, to some extent, capture complex real-world relationships.

2.6.3. Principle Sensitivity Analysis

Following are the sensitivity analysis for the 6 principles. First, we defined the range for each principle's score, setting the score for each principle between 0 and 10 based on the scoring calculation discussed earlier. When conducting the sensitivity analysis for any given principle, we set the scores of the other principle to fixed values. Then, for the principle being considered, we divided the scores into 21 data points, which were input into the model to compute the corresponding results. However, after completing the calculations, we found that the scores obtained from the entropy weight method combined with TOPSIS were nearly fixed. Therefore, we used the scores derived from AHP combined with TOPSIS as the primary reference and plotted the following graph：

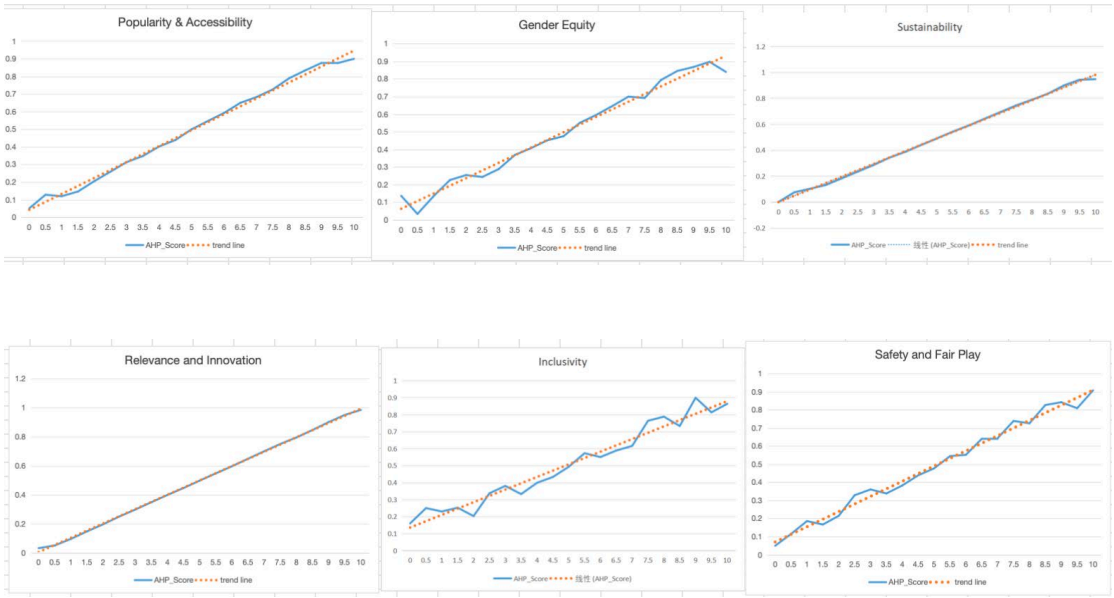


Figure 8: Principle Sensitivity Analysis

Due to the interference of random small perturbations in our code's calculations, in order to avoid factors with same values to have weight of exactly 0, all the graphs are not absolute straight lines. Therefore, we plotted trendlines for each graph to assist in analyzing slopes, and based on this, we can draw the following conclusions:

For all principles, when their scores increase, the final score also increases. However, from the subtle differences in their slopes, the degree and speed at which the score rises will vary. This also demonstrates that our model aligns with real-world situations. For example, when the scores for other criteria are fixed, if a sport scores higher in the criteria of popularity and accessibility, its final score will also be higher. However, if we change the principle whose score is varying to any other criterion, the extent to which the final score increases will change depending on the changes in the weights of the different principle.

2.6.4. Method-Weight Sensitivity Analysis

Finally, since we used weighted average method to calculate the final score, we performed a sensitivity analysis on the weights of TE_Score and TA_Score.

Since the sum of the two weights must equal 1, changes in the weight of TE_Score reflects changes in the weight of TA_Score. In the calculation process, we used the 2024 artistic gymnastics score as the base, dividing the weight into 9 data points, plotting the following graph:

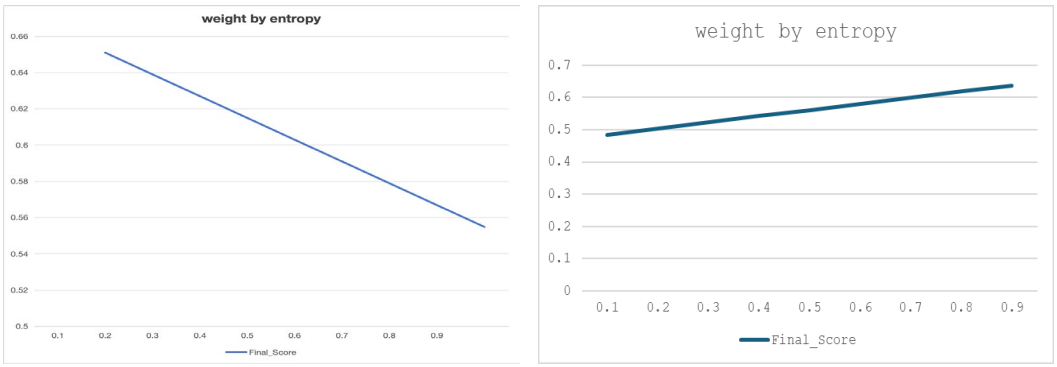


Figure 9: Method-Weight Sensitivity Analysis

From the trend of the left graph, the following conclusions can be drawn:

When the weight of TE_Score is higher, the final score of the SDE is lower. This also shows that our model aligns real data: we selected artistic gymnastics as the scoring object, with a TE_Score of 0.54, and a TA_Score of 0.66. Therefore, when the weight of TE_Score increases, the final score decreases.

From the right graph, opposite scenario shows final score increases with weight.

We can see that the influence of the weights of the two methods on the final score is not fixed, but depends on the relative scores obtained by the SDE in each method. However, as seen from the slopes of the above graph, the sensitivity of the two weights is still relatively high.

3. Strengths and Weaknesses

3.1. Strengths

1. Our model considers 21 variables, making it relatively comprehensive.
2. Our final model integrates three widely used evaluation models. Each of these models has the capability to independently produce a scoring result, while we

selected them based on their respective strengths and ultimately combined them into a single model, demonstrating a degree of innovation.

3. We utilized the entropy weight method to mitigate the subjectivity inherent in the AHP method while simultaneously using the AHP method to offset the data dependency of the entropy weight method. This approach ensures that the results are relatively objective and accurate.

3.2. Weaknesses

1. We assumed that SDEs do not influence each other, which allowed us to ignore complex and difficult-to-quantify factors (such as the similarity and redundancy of events). However, this also means that the factors considered in our model are not comprehensive enough.
2. The two models we used to calculate weights each have their own limitations:
3. The entropy weight method heavily relies on the collected data to calculate factor weights.
    1) However, since we do not have access to authoritative databases and had to search online, our sample data is still somewhat incomplete and inaccurate, which could likely reduce the accuracy of the model.
    2) When using the AHP model for weight calculation, the weights are entirely assigned subjectively, which makes the method highly subjective and could negatively impact the objectivity of our model.
4. Even though we combined the two methods to mitigate these limitations to some extent, it is still impossible to eliminate these shortcomings entirely. Therefore, this remains a significant drawback.
5. The method we used to combine TE_Score and TA_Score is relatively simple. As mentioned in Section 2.6.4, the weight that maximizes the final score can vary for each SDE based on its respective scores. But assigning different weights to each SDE for calculating the final score is clearly impractical and we still chose to assign a fixed weight of 60% for TE_Score and 40% for TA_Score across all SDEs. However, this approach prevents us from maximizing the score for each SDE, introducing another limitation.

4. Letter to the IOC

HIMCM Team #15926
Nov 19th, 2024

To,
Thomas Bach
President of the International Olympic Committee,
Lausanne, Switzerland

Dear Mr. Bach,

First of all, thank you! Knowing the importance of choosing suitable SDEs to the success of an Olympics game, we are very honored to contribute to the evaluation process of them, and are very happy to present our evaluation model now.

As the Olympics promotes global inclusivity and has a significant influence in the sports world, we first need to carefully consider which features of the SDEs to use as factors of our model. As for this, we used a total of 21 factors, following the six principle criteria, and including as many aspects in our model as possible.

Next, we facilitated three widely-applied evaluation models and combined them into our own integrated model. We first adopted two different methods to compute the weight of the 21 factors, the Entropy Weight Method and the Analytic Hierarchy Process (AHP) method, with the former method relying more on objective data while the latter one including more subjectivity, we can obtain two sets of weights to address the mixed subjectivity and objectivity of our data. Then we passed these weights into the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) model to generate two scores for each SDE, and finally calculate one final score using weighted average of the two scores.

After finishing model constructing and testing, we applied it to predict most possible SDEs that will be added in the 2032 and 2036 Olympics. From the outcomes of our model, we concluded that Teqball and League of Legends are most possible to be introduced in the 2032 Olympics, while field handball is likely to be reintroduced. For 2036, baseball is likely to be reintroduced.

It will be a great honor if you considered our model and suggestions. With these SDEs, we hope to maximize the six criteria given by IOC and make the Olympics as successful as possible.

Sincerely,
HIMCM Team #15926

5. References List

[1] Evaluation criteria for events, 8 June 2015, International Olympic Committee
https://stillmed.olympics.com/media/Document%20Library/OlympicOrg/IOC/Who-We-Are/Commissions/Olympic-Programme/Olympic-programme-Host-city-proposal-Evaluation-criteria.pdf?_ga=2.183395748.304873666.1627395939-amp-UglpLEJdCMnJ_sH5bkzn1g
[2] Facts and figures of women in sports, UNWomen
https://www.unwomen.org/en/paris-2024-olympics-new-era-for-women-in-sport/facts-and-figures-women-in-sport
[3] The World's Highest-Paid Athletes, 2024, Forbes
https://www.forbes.com/lists/athletes/?sh=2e9ff7e95b7e
[4] Recognised International Federations, International Olympic Committee
https://olympics.com/ioc/recognised-international-federations
[5] The World Anti-Doping Code, WADA
https://www.wada-ama.org/en/what-we-do/world-anti-doping-code
[6] Audience measurement, Nielsen
https://www.nielsen.com/solutions/audience-measurement/
[7] Sports Industry, KPMG
https://kpmg.com/us/en/industries/sports.html
[8] A whole new ball game: Navigating digital transformation in the sports industry, Deloitte
https://www.deloitte.com/global/en/Industries/tmt/perspectives/gx-digital-transformation-and-future-changes-in-sports-industry.html
[9] The Olympics are part of a push to "green" sports. Here's why that's important, 02 August 2024, UNEP
https://www.unep.org/news-and-stories/story/olympics-are-part-push-green-sports-heres-why-thats-important
[10] World Athletics Sustainability Policy, December 2021, World Athletics
https://worldathletics.org/athletics-better-world/sustainability/world-athletics-sustainability-policy

6. Appendix

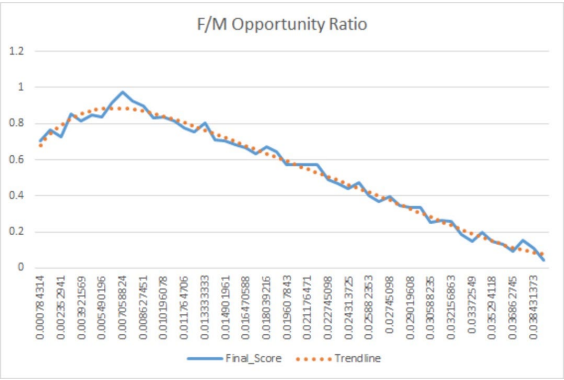


Figure 10: F/M Opportunity Ratio Score